DESCENT DATA FOR A FAMILY OF MORPHISMS IS THE SAME AS DESCENT DATA FOR THEIR COPRODUCT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Fibre products commute with coproducts in the category of schemes. Standard, used at proof lines 8488-8489. Supplies the identification of the fibre products of the coproduct with coproducts of fibre products.
- Descent data relative to a family and relative to a single morphism. Same chapter, the definitions named in the statement at lines 8473-8481. Supplies the two categories being compared.

Lemma 0.1. *Let S be a scheme. Let $\{X_i \to S\}_{i\in I}$ be a family of morphisms with target S. Set $X = \coprod_{i\in I} X_i$, and consider it as an S-scheme. There is a canonical equivalence of categories*

$$\begin{array}{c}\text{category of descent data}\\ \text{relative to the family } \{X_i \to S\}_{i\in I}\end{array} \longrightarrow \begin{array}{c}\text{category of descent data}\\ \text{relative to } X/S\end{array}$$

which maps (V_i, φ_{ij}) to (V, φ) with $V = \coprod_{i\in I} V_i$ and $\varphi = \coprod \varphi_{ij}$.

Proof. Observe that $X \times_S X = \coprod_{ij} X_i \times_S X_j$ and similarly for higher fibre products. Giving a morphism $V \to X$ is exactly the same as giving a family $V_i \to X_i$. And giving a descent datum φ is exactly the same as giving a family φ_{ij}. □

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